

The cover features a complex design with a low-poly geometric background in shades of green and grey. At the top, a network of purple circuit lines with circular nodes extends across the width. In the center, a vibrant pink Protea flower is positioned behind the 'AI' portion of the main title. The title 'SAICSIT' is rendered in a large, bold, dark green sans-serif font, with the 'AI' characters having a white fill and a green outline. To the right of the main title, the year '2026' is displayed in a smaller, solid green font. Below the title, the full name of the conference is written in a smaller, dark green, all-caps sans-serif font. At the bottom, a wide-angle photograph of Cape Town, South Africa, is shown, featuring the city's coastline, the harbor, and the iconic Table Mountain in the background. Overlaid on the top of the photograph is another set of purple circuit lines. The bottom of the cover is dominated by the words 'PROGRAMME BOOK' in a large, bold, light green sans-serif font.

SAICSIT

SOUTH AFRICAN INSTITUTE OF COMPUTER SCIENTISTS
AND INFORMATION TECHNOLOGISTS CONFERENCE

2026

PROGRAMME BOOK

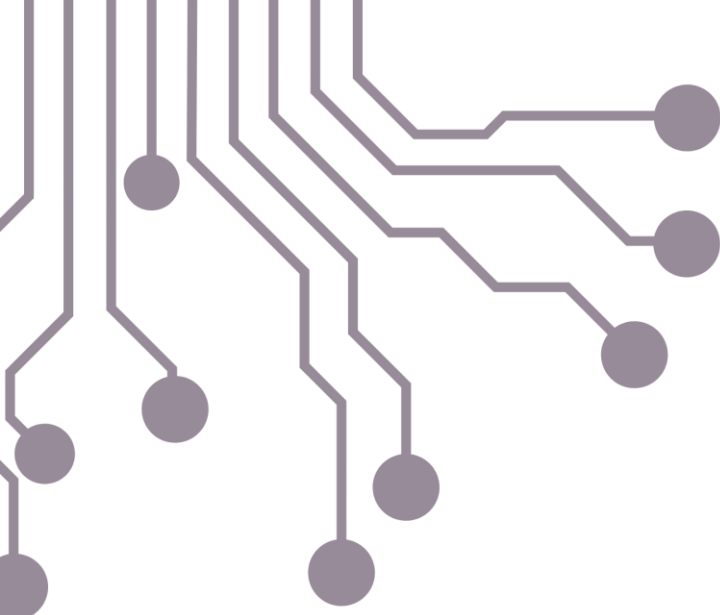


TABLE OF CONTENTS

Message from the SAICSIT 2026
conference chair

SAICSIT 2026 committee

SAICSIT 2026 volunteers

SAICSIT 2026 sponsors

SAICSIT 2026 programme:

- Invited talks
- Camera ready papers info
- Day 1
- Triple Helix workshop
- Postgraduate symposium
- Day 2
- Day 3
- Social events

Africa's greatest resource isn't data

**It's the people who can
transform it into discovery**

**Every algorithm
begins with curiosity**

**Every dataset
holds new possibilities**

**Every breakthrough
starts with collaboration**

NITheCS empowers researchers to turn data into discovery, discovery into innovation, and innovation into solutions to benefit Africa and the world.

Through cutting-edge research, collaboration and capacity development, we are helping shape a future where data science drives scientific breakthroughs, informs better decisions and creates lasting impact.

What would you like to discover?

**Find out more about what NITheCS
can offer you**



www.nithecs.ac.za



NITheCS

National Institute for
Theoretical and Computational Sciences



Connecting
minds



Enabling
discovery



Amplifying
Africa's potential



science, technology
& innovation

Department of
Science, Technology and Innovation
REPUBLIC OF SOUTH AFRICA



SAASTA
South African Agency for Science
and Technology Advancement



MESSAGE FROM THE SAICSIT 2026 CONFERENCE CHAIR

I warmly welcome you to the 47th *South African Institute of Computer Scientists and Information Technologists* (SAICSIT 2026) conference, to be hosted by the School of Information Technology at the University of Cape Town, in Cape Town, South Africa.

This year we had a record number of submissions from all over South Africa and Africa, and we also had one of the most rigorous peer-review processes and lowest acceptance rates in the history of SAICSIT. The SAICSIT 2026 proceedings includes some of the best examples of current Computer Science and Information Systems research from across South Africa, and Africa, spanning a diverse range of topics including: Green AI and Machine Learning, Digital Transformation for Sustainable Development, Innovations in Sustainable Digital Infrastructure, Digital Inclusion and Social Impact, Human-Machine Collaboration for Sustainability, Ethical and Responsible AI Policy, Governance, and Regulation Cybersecurity and Sustainability, AI in Healthcare and Public Wellbeing, Resilient Infrastructure and Disaster Management, Intelligent Transportation and Mobility Sustainable Finance and Fintech, Renewable Energy and Smart Grids, Smart Education and Learning, Analytics Sustainable Supply Chains and Logistics.

I am especially excited about the highly topical theme of Artificial Intelligence (AI) that is framing SAICSIT 2026, and driving the research of many of academic and student delegates participating in this year's SAICSIT. Given the recent increased international interest in AI research and applications, the 2026 conference theme is: ***“The Frontiers of Artificial Intelligence Systems and Sustainable Technologies in Africa”***. A goal of SAICSIT 2026 is thus to engage with research around advancements in Artificial Intelligence technological innovation across Africa, bringing the latest research to the SAICSIT conference forum.

SAICSIT 2026 would never have happened if not for the hard work and support of so many. For this, I sincerely appreciate the support of the School of Information Technology (IT), the University of Cape Town, and the hosting venue: the Graduate School of Business Conference Centre. I am also extremely grateful to the financial support offered by all of our sponsors: the National Institute for Theoretical and Computational Sciences, the School of IT at the University of Cape Town, Amazon and Isazi.ai. I would also like to thank all the members of the programme committee, who have all worked tirelessly to make the SAICSIT 2026 in Cape Town a reality, and to maintain the high academic research standards of the SAICSIT series. Also, special thanks go to all our reviewers for their fastidious efforts in completing reviews in the very short time period between the paper submission deadline and the paper notification date. I would also like to thank our three excellent keynote speakers: Andries Engelbrecht, Bill Tucker, and Kumar Venayagamoorthy for agreeing to deliver fascinating and insightful research perspectives. Finally, many thanks go to all of our session chairs, postgraduate symposium chair, and student volunteers who have helped ensure that SAICSIT 2026 runs smoothly and is a memorable, enjoyable and rewarding experience for all.

I wish every participant of SAICSIT 2026 all the best and am looking forward to seeing you in Cape Town this July.

Geoff Nitschke, General Chair, 47th South African Institute of Computer Scientists and Information Technologists (SAICSIT 2026) Conference



SAICSIT 2026 COMMITTEE



General chair

Geoff Nitschke

University of Cape Town



Technical programme chair

Josiah Chavula

University of Cape Town



Track chair CS

Aslam Safla

University of Cape Town



Track chair CS

Tebogo Bokaba

University of Johannesburg



Track chair IS

Wallace Chigona

University of Cape Town



Track chair IS

Sunet Eybers

University of South Africa



Postgraduate symposium chair

Luzuko Tekeni

University of Cape Town



Sponsorship chair

Luyanda Bheyile

University of Cape Town



Local organising chair

Tharien Potgieter

University of Cape Town



SAICSIT 2026 VOLUNTEERS



Clayton Sikasote

University of Cape Town



Keaton Solomons

University of Cape Town



Tariro Gwandiwa

University of Cape Town



Dennis Kiptanui Too

University of Cape Town



Wilson Kalambo

University of Cape Town



Kyle Inderpersad

University of the Witwatersrand



Nomonde Khalo

University of Cape Town

Thank you to our
volunteers at SAICSIT
2026!





SAICSIT 2026 SPONSORS

Thank you to our sponsors for
making SAICSIT 2026 possible:

NITheCS

National Institute for
Theoretical and Computational Sciences

IT | SCHOOL OF IT

amazon

isazi.ai
— Since 2012 —





BUILD THE FUTURE. TODAY.

About AWS

Amazon Web Services (AWS) is guided by customer obsession, pace of innovation, commitment to operational excellence, and long-term thinking. By democratizing technology for nearly two decades and making cloud computing and generative AI accessible to organizations of every size and industry, AWS has built one of the fastest-growing enterprise technology businesses in history. Millions of customers trust AWS to accelerate innovation, transform their businesses, and shape the future. With the most comprehensive AI capabilities and global infrastructure footprint, AWS empowers builders to turn big ideas into reality.

200+

Cloud Services

32

Geographic Regions

AI

Industry-leading AI

Learn more at

aws.amazon.com

X @AWSNewsroom

SAICSIT 2026 PROGRAMME

INVITED TALKS



ANDRIES ENGELBRECHT

PARTICLE SWARM OPTIMIZATION FOR PREDICTIVE ANALYTICS

Many predictive analytics tasks can be formulated as optimization problems. Examples of such predictive analytics tasks include training of neural networks, training of support vector machines, rule induction, polynomial regression, data clustering, and image segmentation. This talk will present formulations of these tasks as optimization problems, considering both stationary and non-stationary data sources. It will be shown that for some of these tasks, solutions can be formulated using vector representations and others using set-based representations. Particle swarm optimization is a very simple, yet efficient and effective approach to solve a wide range of optimization problems. The talk will proceed to show how particle swarm optimization algorithms can be applied to perform these predictive analytics tasks. Various adaptations of particle swarm optimization will be explored, including approaches to cope with non-stationary data and set-based formulations of particle swarm optimization.





BILL TUCKER

COMMUNITY BASED CO-DESIGN

Technology and social impact became my passion upon starting an academic career at UWC. I emigrated with a South African wife in 1997, after completing a Masters in Computer Science at ASU; working at a spin-off for 6 years during the dot com boom; and honeymooning around the world rock climbing for a year. I started as a technical assistant in 1998 linking ATM and IP networks for the Telkom CoE at UWC; which at the time also included UCT and Stellenbosch. I lectured Networks and Operating Systems, and was later asked to give Human Computer Interaction and ICT4D. Around 2000, however, it became apparent a PhD was required to stay in the game, which I did at UCT whilst working full-time at UWC. Under Edwin Blake and the late Gary Marsden, the PhD led to community-based co-design where Action Research met software development; a cyclical OSI stack equivalent for tech co-design with and for marginalised communities. We learned to build tech tools together to make a real difference in communities and academia. Projects called SignSupport for assistive technology and Deaf communities; and Zenzeleni (a non-profit spin-off) for rural community networks, led to a CBCD trajectory at Stellenbosch. Digital renarration with a local school for Deaf children and the remote Nyae Nyae Village Schools project in Namibia ignited a new passion to nurture pipelines for sustainable community interventions with philanthropic funding. Work with LLMs commenced after being asked by a former UWC PhD student in Pharmacy to code multi-lingual medicine instruction. An even larger project emerged when SU Linguistics asked for help digitising data collection and closing the feedback loop for Early Childhood Development leveraging 4IR tech. MICT SETA provides foundational support across for multi-disciplinary social impact efforts; just as the Telkom CoE did for so many across SA. Multi and interdisciplinary speak to me. I love the mingling of Information Systems and Computer Science at SAICSIT; and hope that relaying my academic journey might inspire you, too, to think about how community-based co-design might help your careers, too.



KUMAR VENAYAGAMOORTHY

THE NEED FOR INNOVATIONS TOWARDS SUSTAINABLE AI

Africa stands at a pivotal moment in its development. Rapid urbanization, expanding electrification, increasing penetration of renewable energy resources, growth in distributed energy systems, and the need for resilient and affordable electricity are creating both significant challenges and unprecedented opportunities. Simultaneously, the modern era of Artificial Intelligence (AI)—driven by machine learning, deep learning, generative AI, foundation models, edge intelligence, and advanced computational capabilities—is redefining how complex infrastructures are designed, operated, and optimized.

This keynote will present a forward-looking perspective on the opportunities of the modern AI era with applications relevant to emerging and developing economies, particularly in Africa. The keynote will also address critical challenges that must be overcome for responsible and effective deployment of AI in mission-critical infrastructures. These include issues of trustworthiness, explainability, robustness, data scarcity and quality, cyber resilience, infrastructure limitations including energy sustainability for AI data centers, workforce preparedness, policy and regulation, and the integration of human expertise with intelligent decision-making systems. Opportunities for AI-enabled innovation that can support universal energy access, improve grid reliability, and foster sustainable economic development across the African continent will be discussed.

Drawing from decades of pioneering research and global experiences in computational intelligence and its applications, this keynote will provide insights into the future of AI-driven energy systems and inspire pathways for research, education, innovation, and international collaboration to build the next generation of sustainable AI.





DARIO FANUCCHI

THE SURVIVOR'S GUIDE TO BEING HUMAN IN THE AI REVOLUTION

I built Isazi on the belief that real-world problems can be transformed by applying mathematics, deep technical skill, and original thinking. We have always been both builders and heavy users of AI, but the last six months have been disruptive even for us. If an AI model can inspect the data, what is the value of a data scientist? If an LLM can write thousands of lines of code, what is the value of an AI engineer? I have watched technology leaders around me respond by downsizing engineering teams, lowering the technical bar, or treating expertise as less important than model access. I think they are wrong. In this talk, I will argue that deep expertise matters now more than ever. AI has changed the nature of technical work, but it has not removed the need for judgement, taste, problem selection, mathematical thinking, domain understanding, evaluation, and system design. If anything, powerful models make these human capabilities more valuable, because they determine whether AI is pointed at problems that matter and whether its outputs can be trusted. I will share Isazi's journey building applied AI systems, including examples from forecasting, education, document automation, and agentic optimisation. I will also unpack my view of human-centric data science in a world of increasingly capable AI models, and what it means to build a real, lasting career if you are starting out in 2026.

CAMERA READY PAPERS INFO

Camera-ready papers for SAICSIT 2026 are available on Google Drive. The papers are organised into folders by conference session for the parallel Computer Science and Information Systems tracks.

You can access the papers here:

<https://drive.google.com/drive/folders/1bNg8vGTZT3aJlsbtmcHxJvsSTBrJ57cr?usp=sharing>

We encourage you to download the papers for the sessions you plan to attend.

IF YOU LIKE MATHS, KEEP READING.

Why Choose Us?

We hire fresh graduates from mathematics, physics, statistics, computer science, and electrical engineering.

You will work alongside people who have shipped AI to production for over a decade.

Your first six months: a real project with a senior mentor, a professor running our internal problem-solving programme and monthly reflections from the founders on what they are working on.

Nobody here uses AI as a buzzword. If you like maths, come use it.

About Us

Isazi means "scientist" in isiZulu. Founded in Johannesburg in 2012 by three Wits mathematicians who believed hard business problems could be solved with maths. Today, our team of 85+ scientists has delivered 175+ projects across banking, healthcare, manufacturing and more, in South Africa, the UAE, the UK, and the USA.

Talk to us

Isazi Origins

A 4-week paid programme for third-years and honours students. You will spend it on a real AI project, working alongside our data scientists and engineers. Apply: send CV & latest academic transcript to gen@isaziconsulting.co.za

Fresh graduate applications open

Data scientists, ML engineers, software engineers.
Scan to apply.





SAICSIT

SOUTH AFRICAN INSTITUTE OF COMPUTER SCIENTISTS
AND INFORMATION TECHNOLOGISTS CONFERENCE

2026

FINAL PROGRAMME

MONDAY 13 JULY 2026

08:00	Registration VENUE: UCT Graduate School of Business (GSB) Conference Centre, Foyer (Ground floor)	
09:00	Welcome by Geoff Nitschke – SAICSIT 2026 General chair and Brandon Collier-Reed – UCT Deputy Vice-Chancellor: Teaching and Learning, Invited talk by Andries Engelbrecht – “Particle Swarm Optimization for Predictive Analytics” VENUE: UCT Graduate School of Business (GSB) Conference Centre, Auditorium	
10:30	 Coffee/Tea 	
	CS: LOGIC, FORMAL METHODS & ALGORITHMS Session chair: Tommie Meyer VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 5 (First floor)	IS: AI IN EDUCATION & DIGITAL LEARNING Session chair: Sunet Eybers VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 4 (First floor)
11:00	C. Hamayobe, T. Meyer & J.P. Haldimann "KLM-Algorithms: An Online Framework for KLM-Style Defeasible Entailment and Explanation"	U. September "The EdRecSys Evaluation Framework: A Design Science Contribution to User-Centric Recommender Systems Theory in Education"
11:20	C. van Wyngaardt, N. Timm & S. Gruner "Formal Schedulability Analysis of Preemptive Periodic Tasks in Real-Time Systems"	O. Olaitan & I. Onwuegbuzie "AI and the Future of Higher Education Assessment: An Analytical Scoping Review"
11:40	R. Carlinsky, N. Timm & S. Rananga "Lightweight Learning-Guided Variable Selection for Modern Satisfiability Solvers"	J. Osakwe & M. Ziezo "Effectiveness of Mobile Intelligent Tutoring Systems in Education"
12:00	S. Jordaan & N. Timm "Nash Equilibrium Strategy Synthesis for Region-Control Multi-Agent Systems via Q-Learning-Guided SMT Solving"	R. Oluwatuyi & M. Densmore "AI Literacy for Young Learners: Experiences and Concerns of Educators in South Africa"
12:20	F.J. Redelinghuys, A.S. Bosman, L. Marshall & N. Timm "Insights into Using Genetic Algorithm with Temporal Logic for Automated Assessment"	C. Mbogho, S. Azeka & J. Edward-Gill "Responsible Computing Beyond the Classroom: Student, Faculty, and Institutional Outcomes Across Eight Kenyan Universities"
12:40	J. Kanjere, M. Ford & B. Mutemula "Simplifying Verifiable Credential Systems: A Hybrid Blockchain-Based Architecture for Credential Issuance and Verification"	K. Aruleba, I.D. Mienye, M. Oloko-Oba et al. "Integrating Green Software Engineering into African Computing Curricula through Systematic Mapping"
13:00	 Lunch 	
14:00	Triple Helix workshop: The inaugural Triple Helix workshop aims to formally launch a collaborative platform that connects computing and information systems researchers with industry leaders and government stakeholders. VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 5 (First floor)	
17:00	 Postgraduate symposium VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 4 (First floor)	
18:30	Welcome reception and Postgraduate symposium poster session VENUE: UCT Graduate School of Business (GSB) Conference Centre, Second floor	



TRIPLE HELIX WORKSHOP

FOSTERING CROSS-SECTOR COLLABORATION FOR DIGITAL INNOVATION
ACADEMIA • INDUSTRY • GOVERNMENT

Call for Participation: 1st Triple Helix Workshop Series, hosted as part of SAICSIT 2026, 13 July 2026

Your voice matters in shaping South Africa's digital future!

The **Triple Helix Workshop Series** brings together academia, industry, and government to collaboratively identify the strategic interventions that should be prioritised over the next five years to accelerate digital transformation, strengthen the digital economy, and develop a globally competitive workforce.

We invite all registered SAICSIT 2026 participants - academic staff, researchers, and students - to join this important conversation alongside senior representatives from government departments and leading organisations across industry. Achieving meaningful and sustainable outcomes requires the active participation of all three sectors. Academia plays a critical role in generating knowledge, developing talent, and ensuring that future skills align with the evolving needs of society and the economy. Your perspectives, expertise, and experiences are essential to informing national priorities and creating practical, collaborative solutions.

Beyond contributing to these important discussions, the workshop series offers significant opportunities for professional engagement and collaboration.

For Academics, the workshops provide an opportunity to:

- Build relationships with government and industry leaders.
- Strengthen partnerships that support research, innovation, and knowledge exchange.
- Gain valuable insights into current and emerging industry skills requirements.
- Inform curriculum renewal to ensure graduates possess relevant and future-focused competencies.
- Strengthen the graduate pipeline by aligning higher education with workforce needs.

For Students, the workshops offer an opportunity to:

- Network directly with employers from both industry and government.
- Gain first-hand insight into the skills and attributes sought by employers.
- Explore career pathways, internships, graduate programmes, and employment opportunities.
- Learn about the diverse and impactful career opportunities available to Information Technology graduates within the South African public sector.
- Develop professional networks that can support future career development.

The challenges and opportunities of digital transformation cannot be addressed by any one sector alone. By bringing together academia, industry, and government, the Triple Helix Workshop Series aims to foster meaningful dialogue, strengthen collaboration, and co-create practical actions that will benefit South Africa's digital ecosystem for years to come.

Be part of the conversation. Share your expertise. Build new partnerships. Help shape the future of South Africa's digital economy.

Questions to be explored during the workshop will be based on the following topics:

1. Aligning Skills with Future Demand
2. Enabling Digital Transformation
3. Building an Inclusive Digital Economy
4. Strengthening Innovation Ecosystems
5. Developing a Sustainable Digital Skills Pipeline

Please let us know if you have any other topics/questions that you would like to explore during this workshop - marie.hattingh@up.ac.za, HKobo@csir.co.za, KCalteaux@csir.co.za

Agenda

- 14:00 - 14:15 Welcome and introduction of discussion questions by co-hosts: Prof Marié Hattingh, University of Pretoria, SAICSIT President and Dr Hlabi Kobo, CSIR
- 14:15 - 15:30 Breakout sessions
- 15:30 - 15:50 Tea/Coffee break
- 15:50 - 16:50 Feedback from breakout sessions, future collaboration agenda - moderated by Lucas Gumbi, CSIR
- 16:50 - 17:00 Closing remarks - Dr Karen Calteaux, CSIR



MONDAY 13 JULY 2026

VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 4 (First floor)

17:00

Talk by DARIO FANUCCHI (Isazi AI):
“The survivor's guide to being human in the AI revolution”



COMPUTER SCIENCE AND INFORMATION SYSTEMS PRESENTATIONS

Session chair: Luzuko Tekeni

17:20

NOMONDE KHALO

“Persona-Aware Medical Text Simplification with Prefix-Tuning and Reinforcement Learning from Human Feedback”

17:23

CLAYTONE SIKASOTE

“Investigating Methods to Mitigate Bias in ASR for Low Resource African Languages”

17:26

UNAYS BHAD

“NextShift: Next Generation NMR Shift Prediction for Carbohydrates via Geometrically Equivariant GNNs”

17:30

KAELA KOKKAS

“AI-Assisted Discovery and Validation of Novel Microbe-Cancer Interactions”

17:33

MAX VAN VEEN

“Synthetic Data Generation for Small Data Orthopedic Radiology”

17:36

SANELE DLAMINI

“A Comparative Analysis of Kubernetes and SLURM for Deploying Multi-User Interactive Scientific Applications: A CARTA Case Study”

17:40

RONALDO NOMBAKUSE

“The Digital Mediation of Informal Mental Health Support for Marginalised Men in Global South: A Gender-Sensitive Approach”

17:43

POPYENI KAUTONDOKWA

“Moving Beyond the Plan: A Practice-Based View of Information Security Strategizing in Developing Economies”

17:46

SHAFIQUE PATEL

“Factors Influencing the Adoption of AI in the Department of Forestry, Fisheries, and Environment”

17:50

TICHAONA CHINYERERE

“A Conceptual Framework for Designing an Adaptive Socio-Technical In-House Electronic Government Procurement System for Growing Economies”

17:53

XING LIU

“Reflexive Constraint in Enterprise Architecture Frameworks: A Diagnostic Construct for Health Information System Governance in Africa”

17:56

TEBOGO LEGODI

“Assessing the Efficacy of Fintech Solutions in Advancing Financial Inclusion for the Financially Underserved in South Africa”

18:00

Expert panel feedback, discussions and closing remarks

COMPUTER SCIENCE PANELISTS:

Jan Buys, Tebogoma Makaba, Deshen Moodley and Dario Fanucchi

INFORMATION SYSTEM PANELISTS:

Bruce Bassett, Siyabonga Mhlongo, Godwin Mandinyenya and Hossana Twinomurinzi

18:30



Welcome reception and Postgraduate symposium poster session

08:30	Registration VENUE: UCT Graduate School of Business (GSB) Conference Centre, Foyer (Ground floor)	
09:00	Invited talk (online) by Kumar Venayagamoorthy – “The Need for Innovations Towards Sustainable AI” VENUE: UCT Graduate School of Business (GSB) Conference Centre, Auditorium	
10:00	 Coffee/Tea 	
	CS: NLP & LARGE LANGUAGE MODELS Session chair: Jan Buys VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 5 (First floor)	IS: CYBERSECURITY: THREATS, PRIVACY & SOCIAL ENGINEERING Session chair: Ntima Mabanza VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 4 (First floor)
10:30	M.Z. Moonsamy, M.M. Dawood, K. Kokkas et al. "Democratizing Discovery: Cost-Efficient Scientific Literature Screening with Open-Weight SLMs"	B. Mutunhu, B. Chipangura & S. Singh "Cybersecurity Risk Perceptions and Trust in Quantified Self Technologies: Evidence from Diabetic Patients in Bulawayo, Zimbabwe"
10:50	N. Alexander, J. Byamugisha, N. Makondo & J. Buys "Lexical Data Augmentation for Medical Text-to-SQL Question Answering"	S. Somerai, Z. Ruhwanya & L. Tekeni "Organizational, Cultural, and Environmental Factors Influencing Information Security Policy Compliance in the South African Banking Sector"
11:10	T. Kolia & S. Levitt "Using Large Language Models for Generating Visualisations from Natural Language Queries Without Data Exposure: A Benchmarking Study"	Z. Mathibela & F. Radebe "Youth Perceptions on Cybersextortion: Implications for Law Enforcement’s Response"
11:30	N. Pather, J. Fouché, S. Mundia, K-G. Technau et al. "From Handwriting to Structured Data: Benchmarking Large Language Models for Handwritten Form Digitisation"	K. Pillay, L. Tekeni & A. Pekane "The Influence of Communication Techniques in Building Trust for Social Engineering Attacks in Higher Education Institutions"
11:50	Short break	
	CS: ML APPLICATIONS IN HEALTHCARE & FINANCE Session chair: Estelle Taylor VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 5 (First floor)	IS: CYBERSECURITY AWARENESS, POLICY & RESILIENCE Session chair: Fani Radebe VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 4 (First floor)
12:00	W.T. Vambe & T.C. Garidzira "Comparative Analysis of Machine Learning Classifiers for Maternal Health Risk Prediction"	G. Mandinyenya & W. Chigona "A Conceptual Analysis of Sustainable Cybersecurity Institutionalisation in South African Rural Schools"
12:20	T.C. Garidzira, W.T. Vambe & C. Gurajena "Oral Guard: Towards an mHealth Application for Oral Cancer Detection"	K. Setati-Legodi, N. Mpekoa & N. Gcaza "Cybersecurity Awareness Gamification for Non-Tech-Savvy Community Members in South Africa"
12:40	T.F. Mapaila & M. Senekane "Enhancing Trustworthiness of Machine Learning-Based Credit Risk Scoring Using Explainability and Uncertainty Quantification"	M. Ngandu, N. Mabanza & G. Mwansa "Towards a Human Centred Protocol for Cybersecurity Information Sharing in South African SMEs"
13:00	 Lunch 	
	CS: ML: EXPLAINABILITY, PREDICTION & VISION Session chair: Deshen Moodley VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 5 (First floor)	IS: DIGITAL AGENCY, AI TRUST & DECISION-MAKING Session chair: Siyabonga Mhlongo VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 4 (First floor)
14:00	J. Muninga, T. Makaba & T. Mokheleli "Explainable Machine Learning for Short-Term Traffic Congestion Prediction"	S. Eybers "Re-thinking Technostress in the Era of Artificial Intelligence (AI): A Socio-Technical System Perspective"
14:20	H.T. Moges & D. Moodley "How Much Does the Spatial Module Really Help? A Systematic Analysis for Long-Term MTS Forecasting"	B. Matandela & L. Tekeni "Understanding the Factors that Influence the Intention of Smartphone Users to Protect against QR Code Phishing"
14:40	S. Mulder & M.C. du Plessis "Evaluating Unsupervised State Encoding in Video Sequences using Disentangled Beta-Variational Autoencoders on Real-World Data"	Y. Buchana & R. Harry "Algorithmic Delegation to Agentic AI Systems in Software Development: Trust, Transparency, and the Capability Ceiling"

15:00	Coffee/Tea	
	CS: DEEP LEARNING, FAIRNESS & ENSEMBLE METHODS Session chair: Zola Mahlaza VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 5 (First floor)	IS: AI IN ORGANISATIONS & PROCESS CONTROL Session chair: Nomusa Mdletshe VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 4 (First floor)
15:30	S. Ncanywa, W. Fourie & D. Brown "Machine Learning in the Archive: CNN-LSTM Generalisation on Archival uhadi Recordings"	A.H. Mazorodze "The Influence of Organisational Culture on User-Acceptance of AI-Powered Decision Support Systems Across Various Organisations"
15:50	I.D. Mienye, G. Obaido, K. Aruleba et al. "FairEnsemble: An Adaptive Weighted Framework for Fairness-Aware Voting in Ensemble Models"	J. Jansen & A. Gerber "AI-Powered Mining: A Taxonomy of Artificial Intelligence within the ISA-95 Process Control"
16:10	B.G. Baleni, A. Usman & C. Nyirenda "A Hybrid VGG16-PCA-Fuzzy Based Framework for the Classification of Morphologically Similar Maize Pests"	E. Veiko & J. Osakwe "Open Data Ecosystems as Catalysts for Smart City Development and Digital Transformation in Africa"
16:30	T.C. Garidzira & S. Ngwenya "A Comparative Positional Information Analysis of BiLSTM, BiGRU and CNN Models for Email Classification"	
19:00	Dinner VENUE: Rooftop on Bree, 170 Bree Street, Cape Town City Centre	

08:30	Registration VENUE: UCT Graduate School of Business (GSB) Conference Centre, Foyer (Ground floor)	
09:00	Invited talk by Bill Tucker – “Community-based co-design” VENUE: UCT Graduate School of Business (GSB) Conference Centre, Auditorium	
10:00	 Coffee/Tea 	
	CS: CYBERSECURITY, NETWORKS & SMART SYSTEMS Session chair: Aslam Safla VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 5 (First floor)	IS: DIGITAL TRUST, TECHNOLOGY ADOPTION & ICT ECOSYSTEMS Session chair: Carolien van den Berg VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 4 (First floor)
10:30	E. Ackerson & J. Chavula "AnDO: A Lightweight Feature Extraction Framework for IDS Modelling in Low-Resource Software-Defined Networks"	V. Pillay, M. Hattingh & J.P. van Deventer "Factors that Influence Individuals' Perception of Digital Trust and Their Sense of Agency"
10:50	J. Roberts & S. Chindipha "A Deep Learning Transformer Approach to Phishing Detection"	M. Sibanda & C. van den Berg "Digital Technology Adoption in Cape Town's Informal Economy: A UTAUT2 Analysis of Barriers and Enablers"
11:10	S. Agenbag, A. Henney & H. Pieterse "Integrating NIST SP 800-101 and ISO/IEC 27000 for Mobile Device Forensics: A Conceptual Framework"	S. Zindela & S. Mhlongo "Strengthening Digital Trust in Public-Sector Cloud Adoption: A Socio-Technical Study of Cloud Security Challenges and Practices"
11:30	S. Makura, H. Venter & C. Ackerman "BlockCoC: A Blockchain-Powered Chain of Custody Model for Cloud-Based Digital Evidence"	J. Osakwe & M. Ziezo "Data Stewardship and Organisational Data Integrity"
11:50	Short break	
12:00	SAICSIT Annual General Meeting (AGM) All delegates, no parallel sessions VENUE: UCT Graduate School of Business (GSB) Conference Centre, Auditorium	
13:00	 Lunch 	

	CS: CYBERSECURITY, NETWORKS & SMART SYSTEMS (cont) Session chair: Stefan Gruner VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 5 (First floor)	IS: DIGITAL TRUST, TECHNOLOGY ADOPTION & ICT ECOSYSTEMS (cont) Session chair: Ricardo Harry VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 4 (First floor)
14:00	R. Tshimange, L. Mamushiane, H. Kobo & L. Mboweni "Performance-Aware Orchestration for Sovereign Government Cloud Infrastructure: An OpenStack-Tacker Research Testbed"	T. Makhosa "5G Network Slicing as an Enabler of Digital Inclusion: Experimental Evaluation for Education and Healthcare Use Cases"
14:20	Z. Gumbi, S. Mtsofso, L. Mamushiane et al. "5G-Enabled Automated Plant Watering System for Small-Farmers Using srsRAN and Open5GS"	N.P. Adegele & K. Pillay "A Contested Terrain Analysis of Innovation Adoption Strategies Among ICT SMMEs in South Africa"
14:40	D. Oladele, M. Sibiya & E. Mnkandla "Carbon-Aware Benchmarking of Lightweight Time-Series Models for Short-Term Solar Power Forecasting in Smart Grids"	T. Moeng & T. Hossana "Augmenting Intelligence in Support Operations: A Postdigital Perspective"
15:00	Short break	
	CS: TEXT MINING, LANGUAGE & CLASSIFICATION Session chair: Marcel Dunaiki VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 5 (First floor)	IS: SOCIAL COMPUTING, HEALTH & SUSTAINABILITY Session chair: Godwin Mandinyanya VENUE: UCT Graduate School of Business (GSB) Conference Centre, VENUE 4 (First floor)
15:10	S.M.S. Patel, O.Y. Okuboyejo & S.O. Olukanmi "Aspect-Based Sentiment Analysis of User-Generated Content on Drug Reviews: Effectiveness, Side Effect and Affordability"	C. Young & M. Hattingh "The Role Social Media Plays in Health Care Information Dissemination for Adults with Hearing Loss"
15:30	Z. Mahlaza, C.M. Keet & T. Magwenzi "On the Ability to Bootstrap Spellcheckers for Nguni Languages via Reusable Model Pipelines"	L. Nyandu & J. Kanjere "Towards Blockchain-Based Carbon Market Integrity: Design of the Mudala Prototype for Carbon Offset Projects"
15:50	T. Elliott, S. Levitt, K. Nixon & M. Bekker "Back-Translation for the Validation of Biomedical Literature Extraction: An Evaluation of Extracted Propositions vs Triples"	S. Sackstein & K. Inderpersad "Systematic Review of Benefits, Challenges and Sustainability Solutions of Electric Vehicles"
16:10	J. du Toit, H. Redelinghuys & M. Dunaiki "Introducing Three New Benchmark Datasets for Hierarchical Text Classification"	H. Andry, H. Tegally, T. de Oliveira & M.U.G. Kraemer "Predicting Human Migration Under Climate Change Using Machine Learning: A Systematic Review"
16:30	 THE END 	






Leading the Future of digital transformation ON THE AFRICAN CONTINENT



Partner with us
<https://bit.ly/3lq1938>



PARTNER WITH THE SCHOOL OF IT

FOR MORE INFO CONTACT: Luyanda Bheyle Manager: School of IT
+27(0)21 650 7001, luyanda.bheyle@uct.ac.za

The School of IT

In 2018, UCT launched the new School of Information Technology in response to the growing needs of the South African and global tech industry, and the increasing demand for university qualifications in the sector.

The 4th industrial revolution is on the horizon and developments such as Artificial Intelligence, Big Data Analysis and Cloud Computing are resulting in technologies becoming part of our daily lives. The Internet is around us all the time with the watches we wear, the phones we use and the way we communicate.

The School of Information Technology at UCT leverages the excellent research and teaching of the Department of Computer Science and the Department of Information Systems to create a new pool of talent, transforming the IT sector and responding to the national agenda for ICT.

SOCIAL EVENTS

SAICSIT 2026 WELCOME RECEPTION

Monday 13 July 2026, 18:30

UCT Graduate School of Business (GSB) Conference Centre
9 Portswood Road, V&A Waterfront, Cape Town

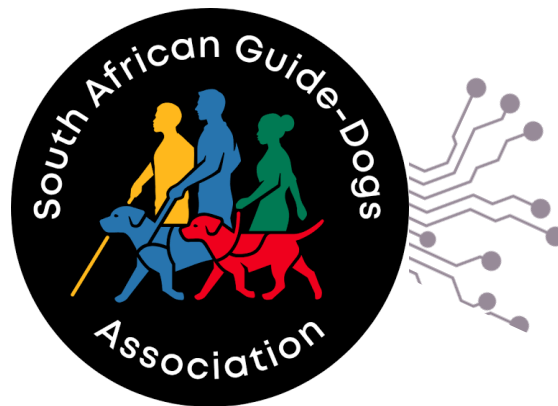
SAICSIT 2026 CONFERENCE DINNER

Tuesday 14 July 2026, 19:00

Rooftop on Bree, 170 Bree Street, Cape Town City Centre

**Conference dinner venue is only accessible by stairs.*

SAICSIT 2026 CONFERENCE CHARITY



The South African Guide-Dogs Association is a non-profit organisation that empowers people with visual impairments, physical disabilities, and autism through expertly trained guide, service, and autism support dogs, as well as mobility training, helping them lead more independent and fulfilling lives.

The South African Guide-Dogs Association will be present at the conference dinner and you are able to donate directly to them during the evening.

